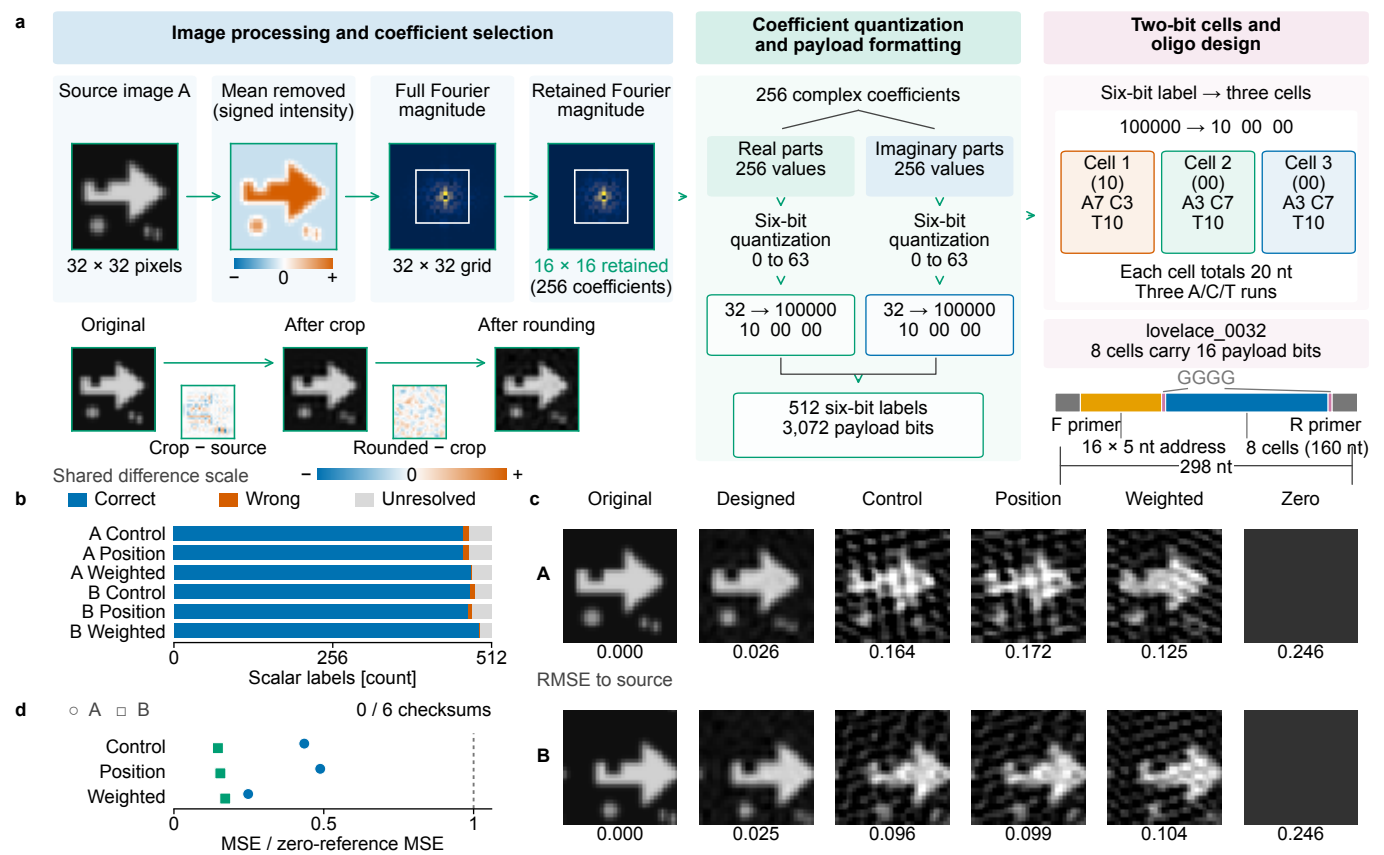

a
Image processing and coefficient selection
Coefficient quantization and payload formatting
Two-bit cells and oligo design
Source image A
Mean removed (signed intensity)
Full Fourier magnitude
Retained Fourier magnitude
32 × 32 pixels
− 0 +
32 × 32 grid
16 × 16 retained
(256 coefficients)
256 complex coefficients
Real parts 256 values
Imaginary parts 256 values
Six-bit quantization 0 to 63
Six-bit quantization 0 to 63
32 → 100000 10 00 00
32 → 100000 10 00 00
512 six-bit labels 3,072 payload bits
Six-bit label → three cells
100000 → 10 00 00
Cell 1 (10) A7 C3 T10
Cell 2 (00) A3 C7 T10
Cell 3 (00) A3 C7 T10
Each cell totals 20 nt
Three A/C/T runs
lovelace_0032
8 cells carry 16 payload bits
GGGG
F primer
R primer
16 × 5 nt address
8 cells (160 nt)
298 nt
Original
After crop
After rounding
Crop − source
Rounded − crop
Shared difference scale
− 0 +
b
Correct
Wrong
Unresolved
A Control
A Position
A Weighted
B Control
B Position
B Weighted
0
256
512
Scalar labels [count]
c
Original
Designed
Control
Position
Weighted
Zero
A
0.000
0.026
0.164
0.172
0.125
0.246
RMSE to source
B
0.000
0.025
0.096
0.099
0.104
0.246
d
○ A □ B
0 / 6 checksums
Control
Position
Weighted
0
0.5
1
MSE / zero-reference MSE